Infosys - Coding 3

A row of N seats is partially occupied. Seat i (1-indexed) is occupied if occ[i-1] is '1', and empty otherwise. You must seat K more people, each in a currently empty seat. Define the discomfort of a seating as the maximum gap between any two consecutively occupied seats. For the purpose of calculating gaps, treat the ends of the row as virtual occupied seats — one at position 0 and another at position N+1. Find the minimum possible maximum gap after placing all K additional people. Note: It is guaranteed that there are at least K empty seats initially.

Case 1:

Input:

7 2 0100100

Output:

2

Explanation:

Initially, occupied seats (including virtual ends) are at positions 0, 2, 5, and 8, creating gaps of 2, 3, and 3. Placing the two additional people at seats 3 and 7 divides the two largest gaps, resulting in final gaps of 2, 1, 2, 2, and 1, with a maximum gap of 2.

Case 2:

Input:

5 0 10010

Output:

3

Explanation:

Since K=0, no additional people are seated. The initial occupied seats at positions 0, 1, 4, and 6 define gaps of 1, 3, and 2, making the maximum gap 3.

Case 3:

Input:

10 0 0101100011

Output:

4

Explanation:

With K=0, occupied seats (with virtual ends) at positions 0, 2, 4, 5, 9, 10, and 11 create gaps of 2, 2, 1, 4, 1, 1 — the maximum being 4.

Input Format

The first line contains an integer, N, denoting the total number of seats. The second line contains an integer, K, denoting the number of additional people to seat. The third line contains a string, occ, representing the initial seat occupancy.

Constraints

1 <= N <= 10^5 1 <= K <= N N <= |occ| <= N

Output Format

The output contains an integer value.

Sample Input 0

```
7
2
0100100
```

Sample Output 0

```
2
```

Explanation 0

Initially, occupied seats (including virtual ends) are at positions 0, 2, 5, and 8, creating gaps of 2, 3, and 3. Placing the two additional people at seats 3 and 7 divides the two largest gaps, resulting in final gaps of 2, 1, 2, 2, and 1, with a maximum gap of 2.

Sample Input 1

```
5
0
10010
```

Sample Output 1

```
3
```

Explanation 1

Since K=0, no additional people are seated. The initial occupied seats at positions 0, 1, 4, and 6 define gaps of 1, 3, and 2, making the maximum gap 3.